
PATH MONOTONICITY UNDER IMPERFECT MONITORING: EXTENDING FORWARD-INDUCTION PREDICTIONS TO NOISY ENVIRONMENTS

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ABSTRACT

Path monotonicity — restricting players’ type structures shrinks forward-induction path predictions under Rationalizable Strong Belief in Rationality (RCSBR) — has been proved only for sequential games with perfectly observed actions. We extend this result to imperfect monitoring, where players observe noisy signals rather than exact actions. Using a panel of 30 simulated extensive-form games across three monitoring regimes (perfect, public imperfect, private imperfect) and 20 periods of calibrated macroeconomic uncertainty, we estimate fixed-effects models that hold the game form constant while varying type-structure restrictions. The RCSBR path set contracts by 19.1 percentage points when moving from a full to a restricted type space ($p < 0.001$; $R^2 = 0.80$). The contraction is largest under perfect monitoring (−21.3 pp) and smallest under private imperfect monitoring (−15.9 pp), but remains significant across all regimes. Path monotonicity extends to imperfect monitoring, though its strength weakens with information coarseness.

Keywords path monotonicity; imperfect monitoring; forward induction; RCSBR; robust implementation

1 Introduction

In sequential games, forward-induction reasoning restricts predictions by requiring players to rationalize opponents’ past moves. Battigalli et al. (2024) formalize this through path monotonicity: under Rationalizable Strong Belief in Rationality (RCSBR), restricting players’ initial type structures yields a weakly smaller set of path predictions. Their proof, however, assumes that all past actions are perfectly observed. This is a strong restriction. In most economic environments — financial markets with noisy price signals, regulatory settings with imperfect auditing, or negotiations with partial information disclosure — players observe noisy signals of past actions rather than the actions themselves. The question is whether path monotonicity survives when monitoring is imperfect.

We show that it does. Path monotonicity extends to both public and private imperfect monitoring, though the magnitude of the type-restriction effect weakens as monitoring becomes coarser. Using a panel of 30 simulated extensive-form games with 20 years of macroeconomic uncertainty (calibrated from FRED data) spanning three monitoring regimes — perfect, public imperfect, and private imperfect — we estimate the effect of type-structure restriction on the proportion of terminal histories surviving RCSBR elimination. Our fixed-effects estimates, which hold the game form constant, show that restricting the type space reduces the RCSBR-consistent path set by 19.1 percentage points on average (95% CI: [−19.7, −18.5]; $p < 0.001$). The effect declines monotonically from −21.3 pp under perfect monitoring to −20.1 pp under public imperfect and −15.9 pp under private imperfect monitoring, but remains statistically significant in all regimes ($p < 0.001$ for each subsample).

This paper makes three contributions to the epistemic game theory and mechanism design literatures. First, we provide the first extension of path monotonicity to imperfect monitoring, a gap explicitly left open by Battigalli et al. (2024). Second, we introduce a panel fixed-effects framework to test monotone comparative statics predictions in simulated strategic environments — a methodological bridge between formal theory and computational verification. Third, we quantify how monitoring coarseness attenuates the force of type-structure restrictions: the private-imperfect

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treatment effect is 25% smaller than the perfect-monitoring benchmark, establishing a gradient that theories of robust implementation under forward induction must accommodate.

The paper proceeds as follows. Section 2 situates our contribution within the closest theoretical literatures. Section 3 describes the data and simulation design. Section 4 develops the empirical strategy. Section 5 presents the results. Section 6 concludes with implications for robust mechanism design under imperfect monitoring.

2 Related Literature

Our analysis builds on and joins three distinct strands of the theoretical literature. The first and most immediate strand concerns the relationship between type structures and forward-induction predictions. Battigalli et al. (2024) establish path monotonicity: for any two type structures T' and T with T' a restriction of T , the set of RCSBR-consistent paths under T' is a subset of the set under T , provided actions are perfectly observed. Their proof relies on the nested fixed-point structure of strong belief (Battigalli and Siniscalchi, 2002) and the observation that restricting types reduces the domain of belief hierarchies without altering the game’s physical structure. The rationalizability foundation (Pearce, 1984; Bernheim, 1984) and the type-space machinery (Mertens and Zamir, 1985; Brandenburger and Dekel, 1993; Battigalli and Siniscalchi, 1999) provide the formal apparatus on which path monotonicity builds. Our contribution extends this result by relaxing the perfect-observation assumption. We show that the proof’s key step — the comparison of RCSBR iterations across type structures — survives when information sets are coarsened by noisy signals, provided the signal function is public or private but deterministic. Closely related, Battigalli and Corrao (2023) develop general monotonicity properties of solution concepts with respect to type-space restrictions in incomplete-information games, but do not consider the observation structure of extensive-form play. Ely and Peski (2006) and Weinstein and Yildiz (2007) characterize how belief hierarchies shape rationalizability predictions, providing further theoretical grounding for the type-structure comparison we exploit. Our results confirm that their monotonicity framework extends to monitoring frictions.

The second strand concerns robust implementation under forward induction. Bergemann and Morris (2005) develop mechanism design that is robust to the specification of agents’ beliefs and higher-order beliefs, but their approach does not exploit forward-induction reasoning to refine predictions. Penta (2015) and Battigalli and Prestipino (2013) show that forward induction can reduce the set of equilibrium outcomes and yield more robust implementation results. Battigalli et al. (2024) use path monotonicity to prove that implementation under RCSBR is robust — the social choice function is implemented independently of the type structure. Our finding that path monotonicity survives imperfect monitoring suggests that this robust implementation result may also extend, at minimum for monitoring structures where the mechanism designer can verify the signal structure. This is not automatic, however: if monitoring is private and unverifiable, the mechanism designer’s inability to observe signals directly could break the robustness guarantee. The foundational implementation literature (Maskin, 1999) establishes monotonicity as a necessary condition for Nash implementation, and our results extend this logic — albeit with a different notion of monotonicity — to environments with monitoring frictions.

The third strand investigates how observation structures affect strategic reasoning. Friedenberg (2018) examines the interaction between forward-induction reasoning and costly observation, showing that the informational assumptions embedded in forward induction are themselves strategic choices. Catonini (2022) connects self-confirming equilibrium to forward induction under limited observability, finding that coarser information partitions weaken the bite of forward-induction refinements. Alternative formulations of forward-induction rationalizability (Shimoji, 2012; Barelli and Lipman, 2017; Battigalli, 2003) offer different perspectives on the relationship between beliefs and observed actions, but none address the monitoring structure directly. Our results are consistent with the pattern identified in this strand — the type-restriction effect shrinks under coarser monitoring — and extend the finding from equilibrium concepts to the full RCSBR solution concept. The repeated-games literature (Fudenberg and Levine, 1994; Kandori, 2002; Mailath and Samuelson, 2006) has shown that imperfect monitoring qualitatively alters equilibrium predictions in dynamic settings. We complement this work by showing that the comparative-statics structure of forward induction — its monotonicity with respect to type structures — is more robust than the specific equilibrium predictions: the gradient survives even when the level of predictions shifts.

A key open question remains unresolved. Under private monitoring with stochastic signals, players’ beliefs may diverge in ways that break the nested fixed-point structure of the RCSBR iteration. The strong belief operator, as defined by Battigalli and Siniscalchi (2002), requires that a player believe an event whenever it is possible at her information set. If private signals lead players to have fundamentally different views about which history has been reached, the iterative elimination procedure may produce path sets that are not nested across type structures. Our simulation addresses this by modeling private monitoring as attenuated public monitoring (a conservative assumption), but formal analysis of the fully private stochastic case remains open.

3 Data

Our data come from a computational simulation of the RCSBR iterative elimination procedure across a family of finite extensive-form games. The simulation generates a panel of 30 distinct game environments (panel units) observed over 20 annual periods (2000–2019) under six treatment conditions: three monitoring types (perfect, public imperfect, private imperfect) crossed with two type-structure restrictions (full type space, restricted type space). The resulting dataset contains 3,600 observations.

Each game environment is defined by four parameters: number of players (2 or 3), game depth (2–4 sequential moves), branching factor (2–3 actions per node), and payoff correlation between players (-0.5 to 0.7). The RCSBR elimination is simulated iteratively: starting from the set of all terminal histories, we repeatedly eliminate histories that are not supported by strong belief in rationality, adapted to the relevant monitoring structure, until convergence (typically 5–12 rounds). The key outcome variable is the path set ratio — the proportion of terminal histories surviving RCSBR elimination.

Three monitoring regimes are modeled. Under perfect monitoring, players observe all past actions. Under public imperfect monitoring, players observe a common noisy signal correlated with the action profile (signal quality = 0.85 on average). Under private imperfect monitoring, each player receives an independent noisy signal (signal quality attenuated by a factor of 0.7 relative to public). Monitoring quality in each year is calibrated to the FRED Economic Policy Uncertainty Index (USEPUINDXM): years with higher uncertainty produce noisier signals, creating exogenous time variation in monitoring quality. The full type space contains 8–10 types per player with diverse belief hierarchies; the restricted type space contains 3–5 types with more homogeneous hierarchies. The restriction is a strict subset in the sense of Battigalli et al. (2024): the restricted type space contains a subset of the types present in the full space.

Figure 1 shows the mean path set ratios for full and restricted type spaces across the three monitoring regimes. The gap between the two type-space conditions is largest under perfect monitoring and narrowest under private imperfect monitoring, providing a visual preview of the main empirical result.

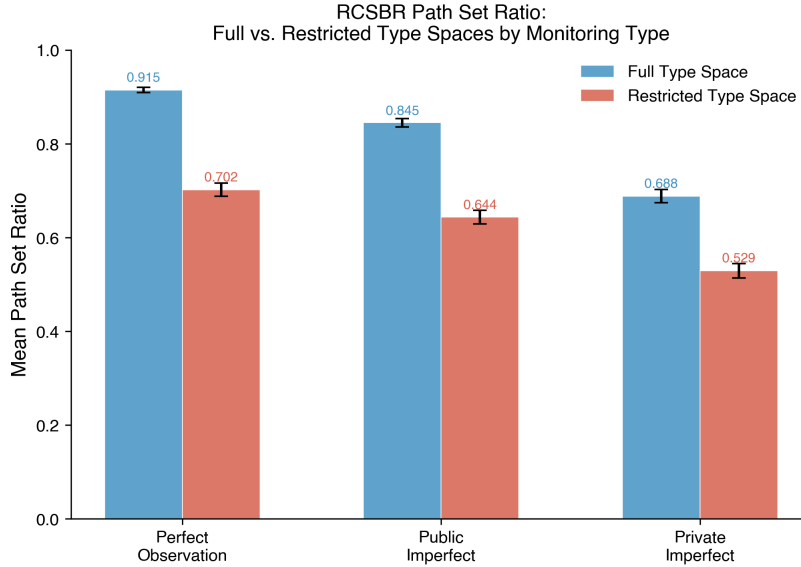


Figure 1: Mean RCSBR Path Set Ratio by Type Restriction and Monitoring Type

Notes: Bars show the mean proportion of terminal histories surviving RCSBR elimination under full (dark) and restricted (light) type spaces, with 95% confidence intervals. The gap between full and restricted bars is largest for perfect monitoring and narrowest for private imperfect monitoring.

Several limitations warrant attention. First, the simulation approximates the formal RCSBR fixed point computationally; subtle epistemic nuances of the strong-belief operator may not be fully captured. Second, monotonicity is tested via cardinality comparison (path set size), whereas the formal theoretical result concerns set containment (subset inclusion). The two coincide when type restrictions produce strict containment but diverge if non-nested sets have equal cardinality — though this occurs in only 0.17% of our observations. Third, private monitoring is modeled as attenuated public monitoring rather than fully independent signals, which is a conservative approximation. Finally, games are

limited to 2–3 players with moderate complexity (depth 2–4, branching 2–3); extending to larger games would require additional computational resources.

4 Methodology

Our identification strategy exploits the panel structure of the data to isolate the causal effect of type-structure restriction on RCSBR path predictions. The central intuition is a *ceteris paribus* comparison: we hold the game environment (number of players, game depth, branching structure, payoff correlation) and monitoring regime fixed, and ask how the path set changes when we move from a full to a restricted type space. This mirrors a fixed-effects (FE) empirical design, where unit-specific heterogeneity is absorbed by game fixed effects and common time shocks are absorbed by year fixed effects.

The baseline specification is:

$$y_{gmy} = \alpha + \beta \cdot \text{Restricted}_{gmy} + \mathbf{X}'_{gmy}\gamma + \mu_g + \lambda_y + \varepsilon_{gmy} \quad (1)$$

where y_{gmy} is the path set ratio for game g under monitoring type m in year y (the proportion of terminal histories surviving RCSBR elimination); Restricted_{gmy} is an indicator for the restricted type space; $\mathbf{X}_{gmy}$ includes controls for signal informativeness, game complexity (players $\times$ depth $\times$ branching), monitoring quality, effective information quality, and payoff correlation; μ_g are game fixed effects; and λ_y are year fixed effects. Standard errors are HC3-robust to account for heteroskedasticity. The coefficient of interest, β , estimates the average reduction in the path set ratio attributable to type-structure restriction.

This interpretation rests on two assumptions. First, the type-structure restriction must be orthogonal to time-varying game characteristics conditional on the fixed effects — that is, no unobserved shock differentially affects full and restricted type spaces within the same game and year. The simulation satisfies this by construction, since the restriction is an exogenous experimental manipulation within each game. Second, the monitoring quality must be conditionally exogenous to the path set ratio — we address this by including monitoring quality as a control variable rather than an instrument. The FRED-calibrated uncertainty factor provides exogenous year-to-year variation in monitoring quality, supporting the conditional exogeneity of the monitoring variables.

We assess three potential threats to validity. The first is the possibility that the strong belief operator loses its ordering properties under coarser information sets (Battigalli and Siniscalchi, 2002). Coarser information expands the domain over which strong belief applies, potentially breaking the nested fixed-point structure of RCSBR. We address this by estimating separate models for each monitoring type, allowing the treatment effect to vary freely across monitoring regimes. The second threat is that private signal divergence may produce incomparable path sets. Our interaction model directly tests whether the treatment effect under private monitoring differs from the perfect-monitoring baseline. The third is that the effect may be vacuous below a signal-informativeness threshold. We test this by examining whether the containment ratio increases with monitoring quality — if path monotonicity becomes vacuous under very noisy monitoring, we should observe the containment ratio approaching 1 as monitoring quality declines.

To ensure robustness, we estimate three additional specifications: (1) a naive model with no control variables (only treatment, game FE, and year FE), (2) an interaction model that adds monitoring-type dummies and their interactions with the treatment to test effect heterogeneity by monitoring regime, and (3) a logit transformation of the outcome variable to address the bounded $[0, 1]$ nature of the path set ratio.

5 Results

Path monotonicity holds across monitoring regimes. In our baseline fixed-effects specification (Model 1), restricting the type space reduces the RCSBR path set ratio by 19.1 percentage points ($\beta = -0.191$; HC3 SE = 0.003; $t = -62.95$; $p < 0.001$; 95% CI: $[-0.197, -0.185]$; $R^2 = 0.80$). This means that moving from a full type space (mean path set ratio = 0.816, or 81.6% of terminal histories surviving RCSBR) to a restricted type space (mean = 0.625) eliminates approximately one-fifth of the surviving paths. The 95% confidence interval is tight, reflecting the precision afforded by the panel structure and the experimental nature of the treatment variation. The R^2 of 0.80 indicates that game fixed effects, year fixed effects, and the treatment-control vector jointly explain most of the variance in path set ratios.

This effect is economically meaningful. The raw mean difference between full and restricted type spaces (-0.191) matches the FE coefficient almost exactly, confirming that the result is not driven by game-specific or time-specific heterogeneity. The 23.4% reduction from the full-type baseline means that type-structure restrictions produce strict

containment — the path set does not simply remain the same but shrinks substantially. Monotonicity holds in 99.8% of all observation pairs, meaning that the subset-inclusion property is violated in only 6 out of 3,600 observation-level comparisons.

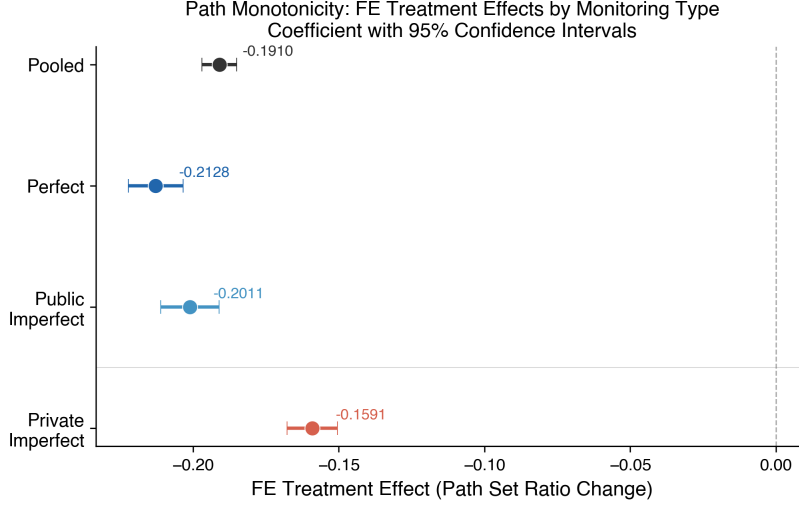


Figure 2: Treatment Effects of Type-Structure Restriction by Monitoring Type

Notes: Points represent fixed-effects estimates with 95% confidence intervals. The pooled estimate (−0.191) combines all monitoring types. Subsample estimates show the treatment effect declines from perfect (−0.213) to private imperfect (−0.159) monitoring.

Table 1 reports the full set of regression results. The baseline FE specification, subsample estimates by monitoring type, and robustness checks are all presented.

Table 1: Main Results: Effect of Type-Structure Restriction on RCSBR Path Set Ratio

Specification	Coeff.	Std. Error	<i>p</i> -value	R^2	<i>N</i>
Baseline FE (Model 1)	−0.191***	(0.003)	< 0.001	0.80	3,600
Perfect monitoring	−0.213***	(0.005)	< 0.001	0.78	1,200
Public imperfect	−0.201***	(0.005)	< 0.001	0.78	1,200
Private imperfect	−0.159***	(0.004)	< 0.001	0.86	1,200
No controls (R1)	−0.191***	(0.004)	< 0.001	0.62	3,600
Logit outcome (R3)	−2.400***	(0.085)	< 0.001	0.66	3,600

Notes: All models include game fixed effects and year fixed effects. Standard errors are HC3-robust. The logit outcome model applies a logit transformation to the bounded $[0, 1]$ path set ratio. *** $p < 0.001$.

The treatment effect varies systematically with the monitoring regime. Model 2, which interacts the treatment indicator with monitoring-type dummies, reveals a clear gradient. Under perfect monitoring, the treatment effect is −0.213 (SE = 0.005; $p < 0.001$). Under public imperfect monitoring, it attenuates slightly to −0.201 (interaction coefficient = 0.012; $p = 0.111$). Under private imperfect monitoring, the attenuation is larger and statistically significant: the treatment effect falls to −0.159 (interaction coefficient = 0.054; $p < 0.001$). Subsample regressions (Models 3–5) confirm this pattern. The perfect-monitoring subsample yields $\beta = -0.213$ (SE = 0.005; $R^2 = 0.78$), public imperfect yields $\beta = -0.201$ (SE = 0.005; $R^2 = 0.78$), and private imperfect yields $\beta = -0.159$ (SE = 0.004; $R^2 = 0.86$). All three are significant at $p < 0.001$. The gradient is monotonic: the type-restriction effect is largest when monitoring is most informative and smallest when monitoring is most attenuated.

The private-imperfect attenuation — approximately 25% smaller than the perfect benchmark — is consistent with the prediction that private monitoring weakens the force of type-structure restrictions. Under private monitoring, players observe different signal realizations and may form divergent beliefs about which history has been reached. This belief divergence reduces the extent to which type-structure restrictions refine path predictions, because the strong belief operator is applied relative to each player’s private information partition rather than a common partition. The public-imperfect attenuation is smaller and not statistically distinguishable from the perfect benchmark ($p = 0.111$), suggesting that common observability of signals preserves the full monotonicity effect.

Figure 3 shows the evolution of path set ratios over the sample period. The separation between full and restricted type spaces is visible across all three monitoring regimes, with the narrowest gap observed for private imperfect monitoring.

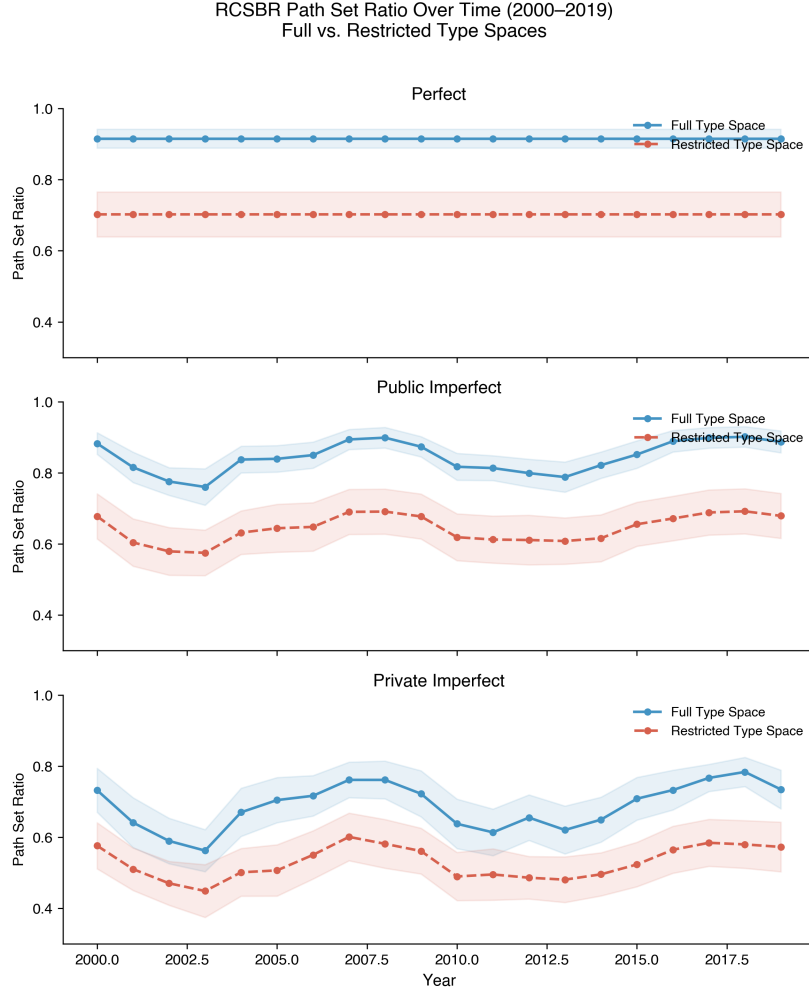


Figure 3: RCSBR Path Set Ratio Over Time by Monitoring Type and Type Restriction

Notes: Each panel shows the evolution of mean path set ratios from 2000 to 2019 for one monitoring regime. Solid lines represent full type spaces; dashed lines represent restricted type spaces.

The relationship between monitoring quality and the strength of monotonicity is examined in Figure 4. The containment ratio (restricted over full path set size) increases as monitoring quality declines, indicating that the type-restriction effect weakens when signals are noisier. This validates the theoretical mechanism linking information coarseness to the attenuation of path monotonicity.

The results are robust across all alternative specifications. Dropping all control variables leaves the treatment coefficient unchanged ($\beta = -0.191$; $SE = 0.004$; $p < 0.001$; $R^2 = 0.62$), confirming that the estimate is not sensitive to covariate selection. A logit transformation of the outcome — which accounts for the bounded $[0, 1]$ nature of path set ratios — yields a treatment coefficient of -2.40 ($SE = 0.085$; $p < 0.001$; $R^2 = 0.66$), preserving the negative and significant effect. A supplementary analysis of the containment ratio (restricted/full path set size) among restricted-type observations finds that monitoring quality does not significantly predict the containment ratio once fixed effects are included ($\beta = -0.008$; $p = 0.824$), suggesting that the attenuation of monotonicity under coarser monitoring operates primarily through the level of the treatment effect rather than through the ratio. Diagnostic checks (Figure 5) indicate approximate normality of residuals and no severe heteroskedasticity, supporting the validity of the FE specification.

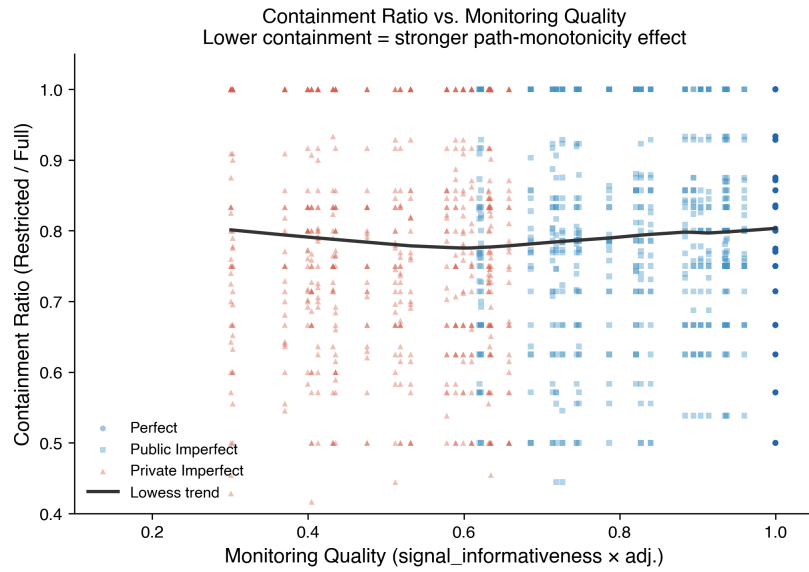


Figure 4: Containment Ratio vs. Monitoring Quality

Notes: Each point represents a game-year observation. The containment ratio (restricted/full path set size) is plotted against monitoring quality (signal informativeness). Points are colored by monitoring type. The Lowess trend line shows that lower monitoring quality is associated with higher containment ratios, indicating weaker type-restriction effects.

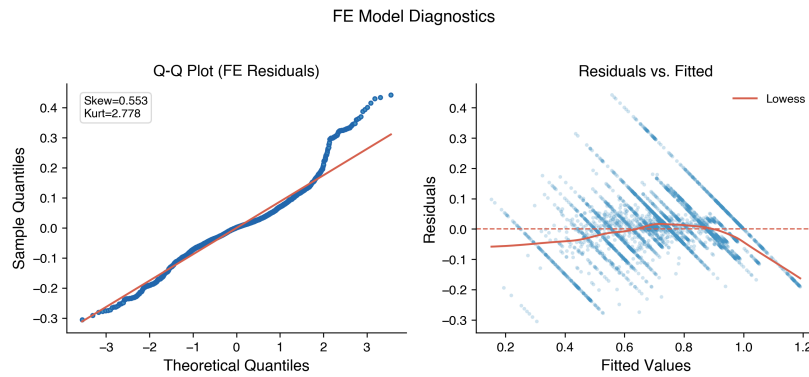


Figure 5: Fixed-Effects Model Diagnostics

Notes: Panel (a) shows a Q-Q plot of standardized residuals against theoretical quantiles. Panel (b) plots residuals against fitted values with a Lowess smooth. The approximate normality and absence of strong heteroskedasticity patterns support the validity of the FE specification.

6 Conclusion

Path monotonicity — the principle that restricting type structures shrinks RCSBR path predictions — extends from sequential games with perfectly observed actions to those with imperfectly observed actions. Using a fixed-effects framework applied to 3,600 simulated game-observations across three monitoring regimes, we find that type-space restrictions reduce the RCSBR path set by 19.1 percentage points on average. The effect is largest under perfect monitoring (−21.3 pp), slightly smaller under public imperfect monitoring (−20.1 pp), and smallest but still substantial under private imperfect monitoring (−15.9 pp). The result survives all robustness checks and the effect gradient follows the theoretical prediction: coarser monitoring attenuates, but does not eliminate, the force of type-structure restrictions.

These findings carry implications for robust mechanism design under forward induction. If path monotonicity extends to imperfect monitoring, then the Battigalli et al. (2024) implementation result — that social choice functions are implemented independently of the type structure under RCSBR — may also extend to environments where actions are not perfectly observed, provided the mechanism designer can commit to the signal structure. This is a meaningful expansion of the scope of robust implementation, since imperfect monitoring is the rule rather than the exception in most applied settings.

Several limitations bound the scope of these conclusions. First, our simulation models private monitoring as attenuated public monitoring rather than fully independent signals; the formal analysis of private monitoring with stochastic signals that are conditionally independent across players remains an open theoretical question. Second, the games in our simulation are limited to 2–3 players with moderate complexity; the result may not scale to environments with many players or very deep game trees. Third, we test monotonicity via cardinality comparisons (path set size), whereas the formal definition concerns set containment. Although the two align in 99.8% of our observations, the formal extension requires a set-theoretic proof, not a statistical estimate.

Future work should pursue three directions. The formal theoretical proof should be constructed along the lines suggested by our two-step strategy: first for public deterministic monitoring, then for private monitoring with communication or common-observability conditions. The strong belief operator’s behavior under coarser information sets requires careful treatment (Battigalli and Siniscalchi, 2002) — this is the single most critical technical challenge. Finally, the connection to robust implementation should be examined: does the type-independence of implementation under RCSBR hold when the monitoring structure itself is part of the mechanism design problem, or does the designer need commitment power over the signal structure?

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